DIGITAL EGYPT PIONEERS INITIATIVE (DEPI) • DATA ANALYSIS TRACK

HR Data Analysis Graduation Project Documentation

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1. PROJECT PLANNING & MANAGEMENT

1.1 Project Idea & Scope

Project Idea: This project develops an interactive Human Resources (HR) Data Analysis Dashboard. As a team of junior data analysts, we take raw HR data stored in Excel, clean and transform it with Power Query, build a relational data model, and write DAX measures in Power BI. The final output is a visual dashboard that helps HR professionals track key employee metrics with ease.

Scope: The project covers importing HR data from Excel, cleaning and transforming the data with Power Query, building a simple relational data model, writing DAX measures (for example, Attrition Rate and Total Headcount), and designing Power BI dashboards.

1.2 Project Goals

- Clean and transform raw HR Excel data into a reliable, analysis-ready format.
- Build an intuitive Power BI dashboard, powered by DAX, that tracks workforce demographics and employee turnover.
- Give stakeholders clear, easy-to-read visuals that support decision-making.

1.3 Team Roles & Responsibilities

Team Member	Role	Responsibilities
Kareem	Team Leader & Coordinator	Data collection, presentation, documentation, and task management.
Arsany	Data Preparation & Modeling	Data cleaning and data modeling.
Abdallah	Dashboard Design	HR Overview page and its DAX, and GitHub submission.
George	Dashboard Design	Demographics page and its DAX.
Youssef	Dashboard & DAX	Departmental Breakdown page and DAX measures and its DAX.

1.4 Timeline & Milestones

Week	Milestone	Owner
Week 1	Requirements gathering, BRD completion, data collection, and initial Excel data preparation	Kareem, Arsany
Week 2	Data cleaning and transformation using Power Query	Arsany
Week 3	Data modeling and DAX calculations in Power BI	Arsany, Youssef
Week 4	Dashboard design (HR Overview, Demographics, Departmental Breakdown), final documentation, presentation, and GitHub submission	Abdallah, George, Youssef, Kareem

2. LITERATURE REVIEW

Traditionally, HR departments have relied on static Excel spreadsheets to track employee data. Excel works well for basic data entry and simple calculations, but it lacks the interactive, drill-down capabilities that modern HR reporting demands. Manual updates are also time-consuming and prone to error, which makes it difficult to keep reports consistent as the data grows.

Our Approach & Competitive Advantage: This project addresses traditional static reporting limitations by taking standard Excel data and elevating it with Power BI. Power Query automates data cleaning, and DAX provides dynamic metrics such as Attrition Rate. The result is an interactive, easy-to-use solution that requires minimal technical knowledge from the end user.

3. REQUIREMENTS GATHERING

3.1 Business Requirements Summary

The purpose of this project is to give the HR department a clear, automated Power BI dashboard that replaces manual Excel reporting. The business needs to quickly identify how many employees are leaving (attrition) and to understand the demographics of the current workforce.

3.2 Stakeholders & Their Needs

- **HR Manager (Primary Stakeholder):** Needs a high-level overview of total headcount, attrition rate, and active employees to report to upper management.
- **Recruitment Team:** Needs to see which departments have the highest turnover so future hiring can be planned effectively.
- **Junior HR Analysts:** Need an easy-to-refresh Power BI file where they only update the underlying Excel sheet, without rewriting complex formulas.

3.3 Functional & 3.4 Non-Functional Requirements

- **Functional Requirements:** Overview Metrics (DAX for Total Headcount, Active Employees, Terminated Employees, Attrition Rate), Demographic Analysis (age distribution, gender ratio, marital status), Departmental Breakdown (employee count and turnover by department), and Interactivity (synced slicers for Year, Department, and Gender).
- **Non-Functional Requirements:** Usability (clean UI/UX suitable for non-technical HR staff), Maintainability (Power Query auto-refresh), and Performance (optimized DAX measures for instant visual response).

4. UI/UX DESIGN STRATEGY & SPECIFICATIONS

The dashboard interface was specifically structured for intuitive, non-technical HR exploration, adhering to modern BI layout standards:

- **Layout & Page Structure:** A three-page main responsive layout (*Workforce Overview*, *Demographics*, *Attrition*) plus a dedicated **Employee Details Drillthrough** page:
  - **Workforce Overview:** Highlights total headcount (10,500), average salary (18,270 EGP), average tenure (5.0 yrs), average performance score (3.33), hires over time trend chart (2013-2022), and headcount by department bar chart.
  - **Demographic Page:** Displays active employees (8,941), female ratio (49.8%), gender pay gap (0.1%), average age (42.2), headcount by age group split by gender, and average salary comparison by gender across departments.
  - **Attrition Overview Page:** Displays attrition rate (9.981%), total leavers (1,048), high performer attrition rate (10.5%), retention rate (90.019%), attrition rate trend over time, and leavers by department.
- **Custom Tooltip Design:** Hovering over any visual element triggers a specialized custom Tooltip card displaying quick-view metrics: Average Monthly Salary, Average Tenure (Years), and Average Performance Score.
- **Drill-Through Detail View (Employee Details Page):** Right-clicking any department or visual enables seamless drill-through to a dedicated granule view showing an operational record-level table: `Employee_ID`, `Job_Title`, `City`, `Hire Date`, `Average Tenure Years`, `Monthly Salary EGP`, and `Performance Score` with a top-left return navigation button.
- **Slicers & Global Navigation:** Top-aligned filter panel with dropdown slicers for `Month`, `Year`, `City`, `Job Title`, and `Department Name` alongside a custom **Clear** button and top navigation pill buttons.

5. PROJECT METHODOLOGY

The team follows a sequential workflow in which each stage feeds the next. Because the Power Query steps are built to be reusable, the dashboard refreshes automatically whenever the source Excel file is updated.

STEP 1	STEP 2	STEP 3	STEP 4	STEP 5	STEP 6
Requirements Gathering	Excel Data Prep	Power Query Transform	Data Modeling	DAX Calculations	Dashboard Design

Core DAX Metrics Engine

TOTAL HEADCOUNT
Count of all recorded employee records

ACTIVE EMPLOYEES
Currently employed workforce total (Is Active = 1)

TERMINATED / LEAVERS
Total employee departures (Voluntary & Involuntary)

ATTRITION RATE %
% turnover ratio calculated via DAX (Leavers / Total)

6. EXPECTED DELIVERABLES

- A cleaned and transformed HR dataset in Power BI semantic model.
- Interactive dashboards covering workforce overview, demographics, and attrition.
- A Power BI data model with DAX measures (Total Headcount, Active Employees, Terminated Employees, Attrition Rate).
- All project files submitted through GitHub along with project documentation and a final presentation.

7. CONCLUSION

This project turns static HR spreadsheets into an interactive Power BI dashboard that lets HR stakeholders understand workforce composition and employee turnover at a glance. By combining Power Query for automated cleaning with DAX for dynamic metrics, the solution remains easy to maintain and accessible to non-technical users, giving the HR department a reliable foundation for reporting and decision-making.